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Natural Disaster Prediction Model on Earthquakes

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June 2021

Certificate

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Abstract

Earthquake prediction has been a challenging research area, where a future occurrence of the devastating catastrophe is predicted. In this work, sixty seismic features are computed through employing seismological concepts, such as Gutenberg-Richter law, seismic rate changes, foreshock frequency, seismic energy release, total recurrence time.

Further, Maximum Relevance and Minimum Redundancy (mRMR) criteria is applied to extract the relevant features. A Support Vector Regressor (SVR) and Hybrid Neural Network (HNN) based classification system is built to obtain the earthquake predictions. HNN is a step wise combination of three different Neural Networks, supported by Enhanced Particle Swarm Optimization (EPSO), to offer weight optimization at each layer.

The newly computed seismic features in combination with SVR-HNN prediction system is applied on Hindukush, Chile and Southern California regions. The obtained numerical results show improved prediction performance for all the considered regions, compared to previous prediction studies.

Acknowledgments

I would like to thank friends and colleagues for the enormous support and valuable discussion, in particular Dr Mehdi Hassan has been very helpful regarding technical discussion of the prediction model. I would also like to thank the teachers and faculty of Air University for the support regarding my project.

Moreover, I would like to express my special thanks of gratitude to my friends who helped me in doing a lot of research and I came to know about so many new things I am really thankful to them.

Secondly, I would also like to thank my parents who helped me a lot in finalizing this project within the limited time frame.

Zarnab Khan

Islamabad, Pakistan June 2021

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Acronyms and Abbreviations

DSA	Data Structure and Algorithms	OOP
Object Oriented Programming	PF	Programming
Fundamentals		
SE	Software Engineering	
SQL	Structured Query Language	
UNESCO	United Nations Educational, Scientific and Cultural Organization	UNICODE
Unique, Universal, and Uniform Character encoding		
XML	Extensible Markup Language	
SVR	Support Vector Regression	
HNN	Hybrid Neural Network	
EPSO	Enhanced particle swarm optimization	
mRMR	Maximum relevancy & Minimum redundancy	

Chapter 1

Introduction

Earthquakes are one of the major catastrophes and their unpredictability causes even more destruction in terms of human life and financial losses. One of the reasons is the lack of technology in accurately monitoring the stress changes, pressure and temperature variations deep beneath the crust through scientific instruments, which eventually results in unavailability of comprehensive data about seismic features. The second probable cause is the gap between seismologists and computer scientist for exploring the various venues of technology to hunt this challenging task.

The main problem is to develop a model that can predict earthquakes using data on seismic features and give accurate results, such as the previous models were not accurate enough in terms of prediction of earthquake on different magnitudes.

Goals and Objectives

□ Goals

Make future earthquake prediction

Make alarming condition to save lives and financial loss.

Make efficient way to predict earthquake.

□ Objectives

To test the training model with data.

To compare the model with existing techniques.

To make earthquake prediction application.

Improve the accuracy

1.1.Scope

- Make future earthquake prediction
- Make alarming condition to save lives and financial loss.
- Make efficient way to predict earthquake.

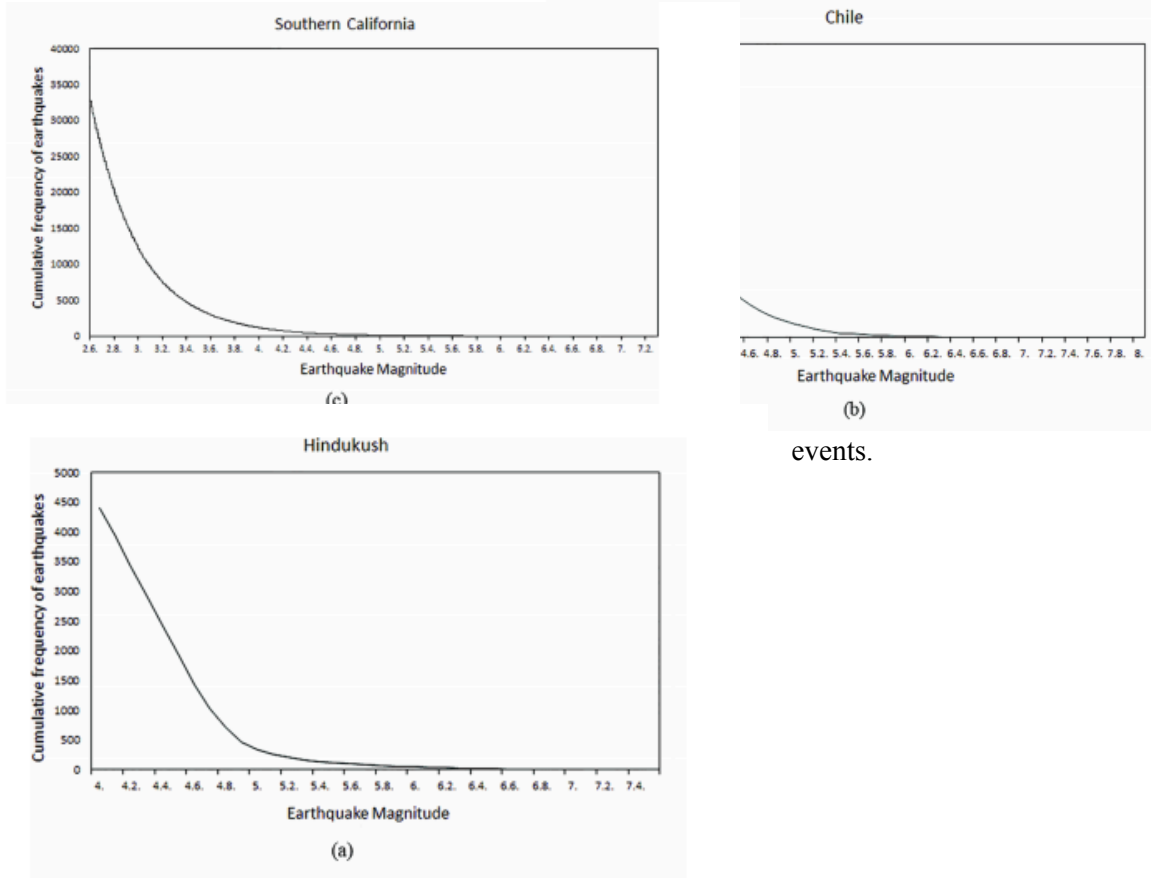
1.2.Project Statement

In this study, research has been carried out for earthquake prediction through interaction of seismology-based earthquake precursors and computer science based computational techniques. Sixty seismic features are computed for Hindukush, Chile and Southern California regions. Separate feature selection is performed for every region through maximum relevance and minimum redundancy (mRMR) approach. The selected features are employed for training of an earthquake prediction model. The prediction model consists of Support Vector Regressor (SVR) followed Hybrid Neural Network. Performance of SVR-HNN over multiple runs of simulation. SVR provides an initial estimation for earthquake prediction which is passed to HNN as an auxiliary predictor in combination with features. Three different neural networks are further employed along with EPSO weight optimization.

Base Paper

Regions selection and feature calculation

3.1 Region selection and earthquake catalog In this study, three different regions, namely Hindukush, Chile, Southern California have been selected for prediction of earthquakes of magnitude 5.0 and above. The same regions selected in the precedent studies are also considered for this research [9, 11, 15]. The advantage of selecting the same regions is that results can be compared in the end, so as to prove the superiority of suggested methodology. Earthquake catalogs of these regions have been obtained from United States Geological Survey (USGS) [28] for the period from January 1980 to December 2016. These catalogs are initially evaluated for cut-off magnitude. Cut-off magnitude corresponds to the earthquake magnitude in the catalog above which catalog is complete and no seismic event is missing. This depends upon the level of instrumentation. Dense instrumentation in a region leads to better completeness of catalog with low cut-off magnitude. The cut-off magnitude for Southern California region is found to be less than 2.6, for Chile it is 3.4 and for Hindukush it is 4.0. The completeness of magnitude for all three regions shows the density of instrumentation in these regions. There are different methodologies proposed in literature for evaluation of cutoff magnitude [29]. In this study, cut-off magnitude is determined through Gutenberg-Richter law. The point where curve deviates from exponential behavior is selected as a cut-off magnitude. All the events reported below cut-off magnitude are removed from the catalog before using for parameter calculation. Earthquake magnitudes and frequency of occurrences for each region is plotted as shown in Fig 1. The curves follow decreasing exponential behavior, which assures that each catalog is complete to its respective cut-off magnitude. After parameter calculation a feature vector is obtained corresponding to every target earthquake (Et). In this study, earthquake prediction problem is designed modeled as a binary classification problem. Every earthquake magnitude is converted to Yes, No (1, 0) through applying threshold on magnitude 5.0. It is too early in this field of research to predict actual magnitudes of future earthquakes; however, endeavors are on the way to predict the categories of future



events.

Parameter calculation Features are the most important part of a classification problem. In this study, features are also referred as seismic parameters. These seismic parameters are calculated mathematically and are based upon well-known geophysical and seismological facts. There are different geophysical and seismic parameters suggested in literature for earthquake prediction studies employing computational intelligence [9, 15, 26]. Discovering new geophysical and seismological facts leading to earthquake prediction is another aspect of earthquake prediction study, which is currently not included in the scope of this research work. Seismic parameters calculated in this research are broadly classified into two main categories with regards of calculation perspective, which are defined as:

- The seismic features whose calculation is not dependent upon any variable parameter are called non-parametric seismic features.
- The seismic features whose calculation is dependent upon any variable parameter, such as a threshold, are called parametric seismic features. The important contribution of this research from the perspective of seismic parameters are:

1. All the available geophysical and seismological features employed for earthquake prediction in contemporary literature are taken into account simultaneously, which has never been done before.

2. Multiple values of parametric seismic features have been calculated based upon different variations of a variable parameter, in order to retain maximum available information about the internal geological state of the ground. All the seismic features are calculated using the 50 seismic events before the event of interest (Et), which is to be predicted using the feature vector. The numbers of features reach to 60 in every instance according to suggested methodology. Later, in order to handle the issues related to curse of dimensionality, the feature selection technique based on Maximum Relevance and Minimum Redundancy (mRMR) is employed to choose the features having maximum relevant and discriminating information. 3.2.1

Non-parametric seismic features.

As the calculation of non-parametric seismic features is not dependent on any variable parameter thus such variables have one possible value for every instance.

- a and b value These values are directly based on well-known geophysical law known as Gutenberg-Richter law. According to this law, number of earthquakes increase exponentially with decreasing magnitude, mathematically shown in Eq 1, where N_i is the total number of seismic events corresponding to Magnitude M_i , b is slope of curve and a is y-intercept.

$$\log N_i = \frac{1}{4} a$$

The values of a and b are calculated numerically through two different methods. Eqs 2 and 3 represent the linear least square regression (lsq), while Eqs 4 and 5 show the maximum likelihood (mlk) method for calculation of a and b values. In earthquake prediction study for Southern California, linear least square regression analysis-based method is proposed [9]. While maximum likelihood method is preferred for earthquake prediction for Chile [15].

$$b_{lsq} = \frac{(n \sum (M_i \log N_i) - \sum M_i \sum \log N_i)}{((\sum M_i)^2 - n \sum M_i^2)}$$

$$a_{lsq} = \sum (\log_{10} N_i + b_{lsq} M_i) / n$$

$$b_{mlk} = \frac{\log_{10} e}{\text{mean}(M) - \min(M)}$$

$$a_{mlk} = \log_{10} N + b_{mlk} \min(M)$$

- Seismic energy release

Seismic energy (dE) keeps releasing from ground in the form of small earthquakes as shown in Eq 6. If the energy release stops, the phenomenon is known as quiescence, which may release in the form of major event. State of quiescence may also lead to the reduction in seismic rate for the region, thereby decreasing b value.

$$dE^{\frac{1}{2}} = \frac{\sum (10^{(11.8+1.5M)})^{\frac{1}{2}}}{T}$$

- Time of n events

Time (T) in days, during which n number seismic events have occurred before Et as shown in Eq 7. In this study n is selected to be 50

$$T = t_n - t_1$$

- Mean Magnitude

Mean magnitude (Mmean) refers to the mean value of n events as shown in Eq 8. Usually the magnitude of seismic events rises before any larger earthquake.

$$M_{mean} = \frac{\sum_i M}{n}$$

- Seismic rate changes Seismic rate change is the overall increase or decrease in the seismic behavior of the region for two difference intervals of time. There are two ways proposed to calculate seismic rate changes. z value shown in Eq 9 measures seismic rate change as proposed by [30], where R1 and R2 correspond the seismic rate for two different intervals. S1 and S2 represent the standard deviation of rate. n1 and n2 show the number of seismic events in both intervals.

$$z = \frac{R_1 - R_2}{\sqrt{\frac{s_1}{n_1} + \frac{s_2}{n_2}}}$$

The other way for seismic rate change calculation is suggested in [31] and given in Eq.10, where, n represents total events in the whole earthquake dataset, t is total time duration and δ is the normalized duration of interest. $M(t, \delta)$ shows the number of events observed, defined using end time t and interval of interest δ . Both z and β values possess opposite signs and are independent from each other.

$$\beta = \frac{M(t, \delta) - n\delta}{\sqrt{n\delta(1 - \delta)}}$$

- Maximum magnitude in last seven days

The maximum magnitude recorded in the days previous to Et is also considered as an important seismic parameter as reported in [12, 15] and represented as x6i. The representation of this parameter is also kept the same as that of literature, so as to maintain better continuity. It is mathematically represented as given in Eq 11.

$$x_{6i} = \max\{M_i\}, \text{ when } t \in [-7, 0)$$

- Probability of earthquake occurrence

The probability of earthquake occurrence of magnitude greater than or equal to 6.0 is also taken as an important seismic feature. It is represented by x_{7i} and calculated through Eq 12. The inclusion of this feature supports the inclusion of Gutenberg-Richter law in an indirect way. The value of x_{7i} is dependent upon b value. Therefore, $blsq$ and $bmlk$ are separately used to calculate x_{7i} , thus giving two different values for this seismic feature.

$$x_{7i} = e^{\frac{-3b_i}{\log e}}$$

- Deviation from Gutenberg-Richter law

It is the deviation η of actual data from the Gutenberg-Richter inverse law as shown in Eq 13. This feature indicates how much actual data follows the inverse law of distribution. Its calculation is dependent upon a and b values, which in turn gives two values for η .

$$\eta = \frac{\sum (\log N - a - bM)^2}{n-1}$$

- Standard deviation of b value

Standard deviation of b value σb is calculated using Eq 14. This feature is parametric because it is based upon the b value, which have two values, therefore adding two values of σb

$$\sigma b = 2.3b^2 \sqrt{\frac{\sum_{i=1}^n (M_i - \text{mean}(M))^2}{n(n-1)}}$$

- Magnitude deficit

Magnitude deficit (M_{def}) is the difference between the maximum observed earthquake magnitude and maximum expected earthquake magnitude (Eq 16). Maximum expected magnitude is calculated through Gutenberg-Richter's law as given in Eq 15. The two sets of a and b values are separately used to calculate M_{def}

$$M_{\max, \text{expected}} = a/b$$

$$M_{def} = M_{\max, \text{actual}} - M_{\max, \text{expected}}$$

- Total recurrence time

It is also known as probabilistic recurrence time (Tr). It is defined as the time between two earthquakes of magnitude greater than or equal to M_0 and calculated using Eq 17. This parameter is another interpretation of Gutenberg-Richter's law. As evident from the statement of inverse law, there will be different value of Tr for every different value of M_0 , which would increase with increasing magnitude. Available literature does not focus on which value of M_0 to be selected in such a scenario therefore Tr is calculated for every M_0 from 4.0 to 6.0 magnitudes following the principle of retaining maximum available information. So, for two sets of a and b values along with varying M_0 adds 42 seismic features to the dataset.

$$T_r = \frac{T}{10^{a-bM}}$$

Support vector regression

SVR is a machine learning technique that learns in a supervised manner. It is first proposed in [36] and implementation is carried out using LIBSVM (A library for Support vector machine) [37]. SVR has wide range of applications for both classification as well as regression problems. The model generated through training of SVR, gives predictions about an estimated earthquake magnitude corresponding to the feature vectors. SVR then imparts its knowledge of predicted earthquake magnitudes to HNN in next step through auxiliary predictions, to be used as a part of feature set. Experiments have proven that the auxiliary predictions from SVR when used in combination with features, adds a distinctive classification capability to the prediction model.

Enhanced Particle Swarm Optimization

There are different nature-inspired optimization algorithms in literature [38–44] however, EPSO has been employed in this study for weight optimization of ANN. EPSO is an evolutionary algorithm. The idea of particle swarm optimization (PSO) is given in [45]. In this optimization methodology, exploration is carried out for finding the best possible solution or position in the search space, like a bird or an organism searches for food. Different factors affect the hunt for best possible solution, like current position and velocity. The record of best local position, global position and worst global position is also kept for generating optimized solution. EPSO is also a well-considered an optimization methodology, which is being used in different application fields and explained in detail in [32].

Hybrid Neural Networks

The next step of prediction model is to train hybrid neural network model. This step is the combination of three different ANNs along with optimization support extended from EPSO. The training is carried out through 1st ANN and weights are then passed to EPSO for further optimization. In case, if ANN is stuck in local minima, EPSO has the capability to guide it out from this situation. The optimization measure for EPSO is set to Matthews Correlation Coefficient (MCC). If the ANN has already learnt the best possible relation between features and earthquakes, EPSO would return the same weight matrix. Thus, it can be said that EPSO is included in this methodology, to save ANNs from being trapped in local minima. The training of this step is carried out for a binary classification problem, in which intention is to predict the earthquakes of magnitude 5.0 and above. Initially the dataset along with auxiliary prediction from SVR is passed to Levenberg-Marquardt neural network. The network trains in back propagation manner, where error at the output layer is back propagated in every

epoch. When the training stops, the weight matrices are passed to EPSO along with training features and targets. EPSO optimizes the weights in terms of MCC and returns the optimized weights back to ANN. The predictions from optimized weights of EPSO are taken as auxiliary predictions in place of SVR-output to be used along with features for next ANN.

BFGS quasi-Newton backpropagation neural network (BFGSNN) is initialized with the weights of previously learnt NN. In this way, the already learnt information is transferred to the next NN along with auxiliary predictors. The network is trained similarly and the weights along with training data are passed to EPSO for optimization via MCC. The optimized weights are then used to initialize Bayesian Regularization Neural Network (BRNN) and training data is passed in combination with BFGSNN auxiliary predictors. EPSO is again employed to optimize the weights of NN. The pseudo-code is also included here for the consideration of the reviewer:

Step by step procedure of SVR-HNN Model:

```

Inputs: T, N, RF, SVR, HNN, EPSO
[T = Training Dataset
N = Numbers of layers of Neural Network
RF = Relevant Features
SVR = Support Vector Regresser
NN = Neural Network
EPSO = Enhanced Particle Swarm Optimization
HNN = Hybrid Neural Network (Combined package of NN+EPSO)]
RF = mRMR (T)
[SVR_Model, SVR_Predictions] = SVR[RF]
for j = 1 to 3 ('j' indicates neural network and EPSO layer)
    if (j == 1) (First NN takes SVR_Predictions as auxiliary input)
        [NNj_Model, NNj_Predictions] = NNj [RF, SVR_Predictions]
        (EPSO optimizes NN, if it traps in local minima)
        [NNj_Model, NNj_Predictions] = EPSOj[RF, NNj_Model,
SVR_Predictions]
    else
        (Predictions of previous NN, becomes input to next NN)
        [NNj_Model, NNj_Predictions] = NNj [RF, NNj-1_Predictions]
        [NNj_Model, NNj_Predictions] = EPSOj[RF, NNj_Model, NNj-1-
_1_Predictions]
    endif
Endfor
SVR_HNN_Model = NN3_Model (The model obtained after SVR and all lay-
ers of NN supported by EPSO)
Performance Evaluation = SVR_HNN_Model[Test Dataset, Actual Labels]

```

In this work, the output of SVR is treated as its own opinion about the earthquake occurrence. Thus, in a bid to improve the performance of earthquake prediction a combination of three ANNs and EPSO, called as Hybrid Neural Network (HNN) is formulated. Seismic Features are passed to HNN for yielding the predictions. At this point of stage, the results obtained through SVR are considered as auxiliary predictions and passed on to HNN, along with other features. The opinion of SVR would also hold importance in terms of discriminating power between earthquake and non-earthquake occurrences. Therefore, when coupled with dataset of other seismic features, it highly improves the discriminating power of earthquake prediction, resulting in improved prediction performance of HNN. In HNN, the weights and outputs of each ANN is passed to the next layer of ANN, to make the consequent layer start learning ahead of a certain point. Thus, the feedback of SVR or ANN at a specific layer enhance the learning of HNN, leading to the improved earthquake prediction. The role of EPSO is to rescue ANN, when it is trapped in local minima through optimizing the weights. However, in case when ANN is performing well, the EPSO refrains from updating the weight. The model is trained using 70% of feature instances and later independent testing is performed on unseen 30% of data. In the training data two-third part is used for training the algorithm while one-third is used for validation of model.

The evaluation of model is performed on unseen feature instances through considering well-known evaluation measures.

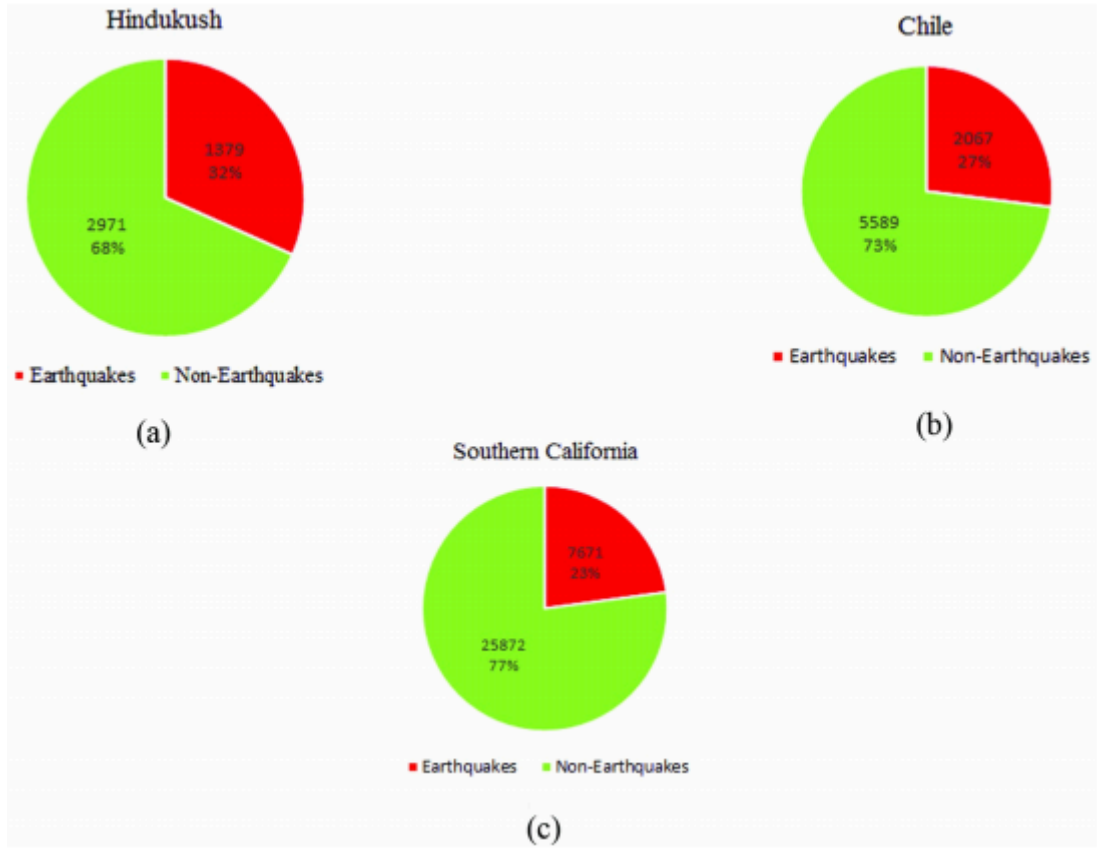
Results and discussion

There are total 7656 feature instances for Chile out of which 2067 correspond to “Yes” earthquake while 5589 correspond to “No” earthquake. The total instances for Southern California are 33543 out of which 7671 belong to “Yes” earthquake while 25872 belong to “No” earthquake.

Similarly, for Hindukush dataset, out of 4350 instances 1379 correspond to “Yes” while 2971 correspond to “No”. Therefore, the data distributions of the considered regions are highly imbalanced.

Performance evaluation criteria

There are well-known performance metrics available for evaluation of binary classification results, such as sensitivity, specificity, positive predictive value or precision (P1), negative predictive value (P2), Matthews correlation coefficient (MCC) and R score. Whenever, predictions are made through an algorithm, it yields four categories of outputs, namely true positive (TP), false positive (FP), true negative (TN) and false negative (FN). TP and TN are the correct predictions made by the prediction model while FP and FN are the wrong predictions of the algorithm. Sensitivity (Sn) is the rate of true positive predictions out of all positive instances while specificity (Sp) is the rate of true negative predictions out among all negative instances. P1 is the ratio of correct positive predictions out of all the positive predictions made by prediction model, whereas P0 refers to the correct negative predictions made by algorithm out of all



negative predictions. P1 inversely relates to false alarms, which refers higher P1 means lesser false alarms and vice versa. Similarly, R score and MCC are also proposed as a balanced measure for binary classification evaluation. These are calculated using all four basic measures (TP, FP, TN, FN) and vary between -1 and +1. The values approaching +1 correspond to perfect classification, while 0 refers to total random behavior of prediction algorithm and -1 relates to opposite behavior of classification model. These two performance measures can be considered as a benchmark measure for drawing comparison, because the four types of basic measures are also incorporated in them as shown in Eqs 22 and 23. Accuracy is a very general criterion for evaluation, used in many perspectives. It states the overall accurate predictions made by the algorithm. The purpose of analyzing the same results through these mentioned criteria is that each performance metric highlights certain aspect of results. Therefore, the purpose is to highlight all the merits and demerits of the results obtained through proposed prediction models.

$$S_n = \frac{TP}{TP + FN}$$

$$S_p = \frac{TN}{TN + FP}$$

$$P_1 = \frac{TP}{TP + FP}$$

$$P_0 = \frac{TN}{TN + FN}$$

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

$$R = \frac{(TP \times TN) - (FP \times FN)}{(TP + FN)(FP + TN)}$$

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Earthquake prediction results SVR-HNN based model is separately trained for all three regions, using 70% of all the datasets. Prediction results are evaluated for all the regions. Accuracy is used for performance evaluation in many aspects. But in unbalanced classification problems, it may not be the best choice to only use accuracy for evaluation. When in a dataset, one class is in abundance and other is less, it is said to be unbalanced dataset. For example, in a dataset of 100 instances, if only 10 instances correspond to earthquake occurrence while rest of 90 belong to no-earthquake then a prediction algorithm with minimal prediction capability predicts all of them as no-earthquakes. In this scenario, the algorithm has no knowledge of predictability but its accuracy would still be 90%, therefore misleading the overall capability of prediction model. However, MCC and R score would yield 0 value for this scenario, thereby giving better insights into the competence of prediction model. This is the reason different aspects of a prediction model is evaluated for better analysis. Earthquake prediction is highly delicate issue and false alarms may lead to financial loss and cause panic, therefore, cannot be tolerated. A prediction model with even less than 50% sensitivity but better P1 is preferable over another prediction model having around 90% sensitivity but lesser P1. Considering the fact that there exists no earthquake prediction system till date.

Thus, results obtained through SVR-HNN model are commendable.

Predictions for Hindukush region.

Asim et al. [11] carried out earthquake prediction studies for Hindukush region where considerable prediction results are obtained through different machine learning techniques. Pattern recognition neural networks (PRNN) yields better results than other discussed methods, therefore PRNN is chosen in this work, for comparison with SVR-HNN based prediction methodology. Table 1 shows that SVR-HNN based prediction results are outperforming PRNN based predictions by wide margin in all aspects, except sensitivity. The value of MCC has been improved considerably from 0.33 to 0.6 and R Score from 0.27 to 0.58. Improvement is also observed in P1 from 61% to above 75%, thus improving false alarm generation for Hindukush region from 39% to less than 25%. Decreased sensitivity is acceptable with notable improvement in P1, MCC, R score and accuracy. A model with increased sensitivity may sensitize false earthquakes, leading to the generation of false alarms. Therefore, a model robust towards false alarms may have less sensitivity, which is acceptable given the other performance evaluation criteria have improved.

Predictions for Chile region.

The results are even more improved for Chile region through SVR-HNN based prediction model. Previously, Reyes et al. [15] carried out earthquake prediction for four Chilean regions through applying different machine learning techniques with ANN achieving the best results. The ANN based results for all the Chilean regions are averaged (average of TPs, FPs, TNs, and FNs) in this study for drawing comparison with SVR-ANN based results. The proposed SVR-HNN approach has underperformed the ANN based Chilean results in the all aspects by considerable margins, except marginal difference in Specificity as evident from Table 2. False alarm generation reduced from 39% to less than 27% along with 5% increase in accuracy as well. The considerable difference can be observed in MCC and R score which increased from 0.39 to 0.61 and 0.34 to 0.60.

Predictions for Southern California region.

Southern California region is considered earlier for earthquake prediction using ANN by Panakkat and Adeli [9]. Recurrent Neural Network (RNN) is reported to have produced better results in terms of R Score. The results were evaluated in terms of false alarms ratio (FAR), Probability of detection (POD) or sensitivity, frequency bias (FB) and R score. To discuss the results through the same evaluation criteria, the values of basic performance parameters (TP, TN, FP and FN) are calculated through set of four equations of FAR, POD, FB and R Score. After calculating basic performance parameters, other evaluation criteria are calculated and given in Table 3. The results generated through SVR-HNN based methodology are better than RNN based results of [9]. False alarm generation are decreased considerably from 29% to less than 7%. A noteworthy increase in MCC and R score is also observed from 0.51 to 0.722 and .051 to 0.62, respectively.

Hence proving the SVR-HNN based prediction methodology better than already available prediction models. SVR-HNN is outperforming previous prediction models because it is a multilayer model with every layer adding to its robustness. SVR provides initial estimation of the earthquake predictions, which is further refined by three different ANNs and EPSO supporting it through optimization.

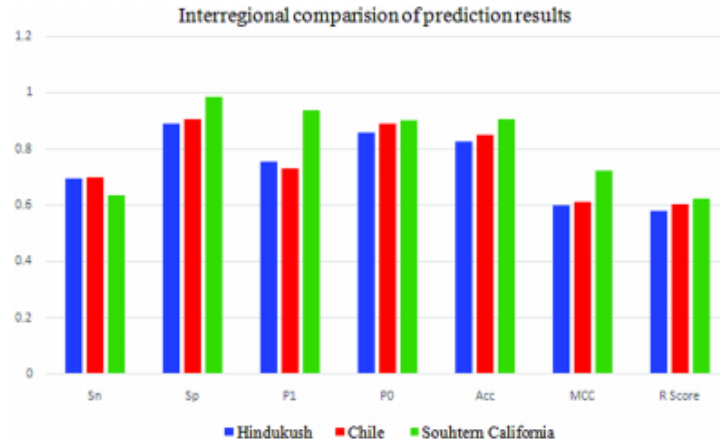
Interregional comparison of earthquake prediction.

SVR-HNN based prediction model has shown improved results for Southern California as compared to the other two regions with 0.722, 0.623 and 90.6% of MCC, R score and accuracy, respectively. While Chilean region is holding second best position with MCC of 0.613, R score of 0.603 and accuracy of 84.9%. The SVR-HNN based proposed model has shown least results for the Hindukush region with MCC, R score and accuracy of 0.6, 0.58 and 82.7%, respectively. Fig 5 graphically compares the prediction results for all the three regions, scaled between 0 and 1. This inter-region results comparison has led to the practical visualization of the fact that lower cut-off magnitude and better completeness of earthquake catalog would lead to the improved results for earthquake prediction. The completeness magnitude of Southern California is taken as 2.6, for Chile its 3.4 and for Hindukush its 4.0.

Hence demonstrated that earthquake prediction results are inversely related to the cut-off magnitude. In other words, it can be inferred that dense instrumentation for earthquake monitoring plays key role for better earthquake prediction.

The overall achievements of this study are presented as:

1. SVR-HNN based prediction methodology generated considerably improved results for Hindukush, Chile and Southern California. It has exceedingly outperformed other available predictions results in literature.
2. Inter-regional comparison of prediction results shows that the region with better maintained earthquake catalog having low cut-off magnitude is capable of generating better prediction results. It highlights the need of better instrumentation for earthquake catalog maintenance.

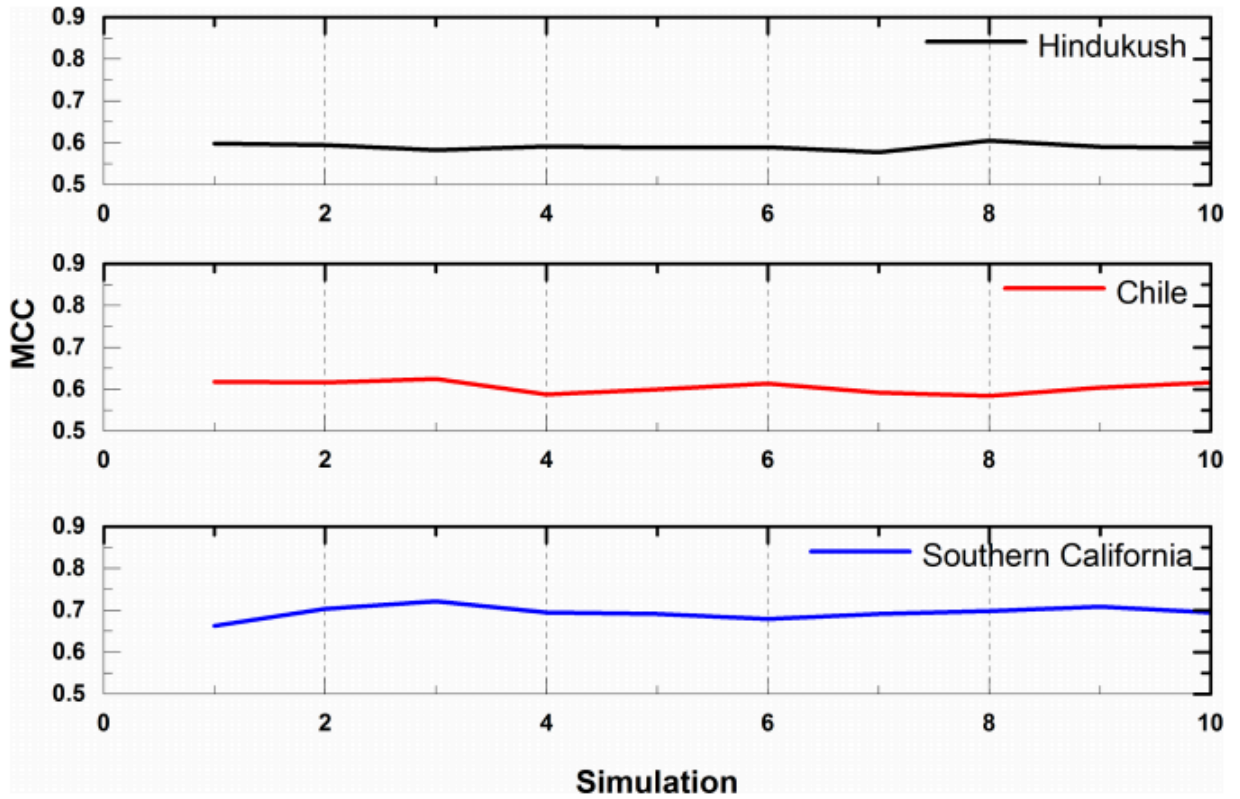


Comparison between individual techniques and SVR-HNN.

In order to prove the superiority of SVR-HNN model, the earthquake prediction performance is also computed separately using SVR and HNN. An independent arrangement has been made to get the earthquake prediction results for SVR and HNN. The performance of SVR and HNN is compared with the combine results of both (SVR-HNN). A notable difference between the performances of individual techniques and their combination can be observed. MCC of 0.43 and 0.41 is obtained for Hindukush region using SVR and HNN, respectively. The combination of two techniques by including SVR as auxiliary predictor for HNN improves MCC to 0.58. A similar trend in improvement is also observed for other performance measures, as shown in Table 4, for the three considered regions. Thereby, proving the superiority of SVR-HNN over their separate counter parts. The role of EPSO is to optimize the weights of neural networks. Artificial neural networks have tendency of getting trapped in local minima. If such a situation happens during the training of a neural network, EPSO helps ANN in escaping local minima and guides in towards optimum solution. But this is not the case every time that ANN is trapped in local minima. It may happen occasionally. If ANN is already performing well and learning in the right direction, the inclusion of EPSO does not affect the weights of ANN in such a situation. The term HNN stands for combination of three different neural networks coupled with EPSO. Sometimes HNN without EPSO would show the similar results as compare to HNN with EPSO, however, on some other training occasions, it may show lesser performance.

Table 4. Performance comparison of SVR, HNN with SVR-HNN.

Region	Hindukush			Chile			Southern California		
Criteria	SVR	HNN	SVR-HNN	SVR	HNN	SVR-HNN	SVR	HNN	SVR-HNN
S _n (%)	50.0	0.541	69.9	43.7	54.7	69.8	55.8	55.0	63.5
S _p (%)	89.0	85.4	89.1	93.6	92.4	90.5	98.1	97.2	98.7
P ₁ (%)	69.2	64.0	75.4	72.3	73.8	73.2	89.9	86.8	93.8
P ₀ (%)	78.3	79.4	85.9	81.4	83.9	89.0	87.9	87.9	90.0
Acc (%)	76.1	75.2	82.7	79.9	81.8	84.9	88.2	87.7	90.6
MCC	0.43	0.41	0.6	0.44	0.52	0.613	0.646	0.62	0.722
R Score	0.39	0.39	0.58	0.37	0.47	0.603	0.537	0.52	0.623



Performance stability.

The performance of SVR, HNN and EPSO are dependent on respective parameter selection. Thus, the collective performance obtained in SVR-HNN model is governed through selection of appropriate parameter values. The extensive experimentation is performed to empirically select the values for parameters, which obtain good results for each of the used algorithms (SVR, HNN, and EPSO). The proposed model has shown consistent performance for the three regions, which approves the appropriate selection of values for parameters. The ten simulation runs of SVR-HNN model show little variation, which strengthens the claim of appropriate selection of parametric values, leading to a stable earthquake prediction performance in three regions.

Chapter 2

Literature Review

Numerous researchers have performed earthquake studies from prediction and forecasting perspectives through different angles. During earthquake preparation process beneath the surface, different geophysical and seismological processes occur. These happenings below the surface are believed to cause changes in sub-soil emission, vertical electric field and ionosphere. These precursory changes are studied and mapped retrospectively with major earthquakes [4, 5]. Earthquake prediction is also studied through observing behavioral changes in animals [16]. The animal behavioral study is carried out using motion-triggered cameras at Yanachaga National Park, Peru. The decline in animal activity has been observed prior to Contamana earthquake of magnitude 7.0.

Earthquake magnitude prediction for northern Red Sea area is carried out in [22]. The methodology is based on the feature's extraction from the past available earthquake records followed by feedforward neural network. These features include sequence number of past earthquakes, respective locations, magnitude and depth. The similar kind of features have also been used for earthquake prediction for Pakistan region using BAT-ANN algorithm [23]. These features do not involve any seismological facts and laws, rather the direct modelling of earthquake sequence number, magnitude, depth and location with the future earthquakes is proposed

Last et al. [10] performed earthquake prediction for Israel and its neighboring countries. The used earthquake data of past years has been treated to first clean foreshocks and aftershocks and then seismic parameters are calculated. The computed parameters are then employed for prediction in combination with Multi-Objective Info-Fuzzy Network (M-IFN) algorithm. The proposed prediction system is capable of predicting maximum earthquake magnitude and total number of seismic events for next year.

All the aforementioned methodologies study the earthquake prediction, while focusing only one region. The prediction models are not applied and tested to other earthquake prone regions and no comparisons are also carried out with the results of other research studies. In this research, the prediction model is applied to more than one region and comparisons are also drawn with the results, available in literature.

Chapter 3

Requirement Specifications

- Existing System

Numerous researchers have performed earthquake studies from prediction and forecasting perspectives through different angles. During earthquake preparation process beneath the surface, different geophysical and seismological processes occur. These happenings below the surface are believed to cause changes in sub-soil emission, vertical electric field and ionosphere. These precursory changes are studied and mapped retrospectively with major earthquakes [4, 5]. Earthquake prediction is also studied through observing behavioral changes in animals [16]. The animal behavioral study is carried out using motion-triggered cameras at Yanachaga National Park, Peru. The decline in animal activity has been observed prior to Contamana earthquake of magnitude 7.0 in 2011. However, the focus of this research is to study earthquake prediction through computational intelligence and machine learning based methods. Algorithm M8 which aims to forecast the earthquake of magnitude 8.0 and above tested successfully at the different regions of the globe along with increased efficiency of intermediate term earthquake forecasting. The study analyzes the earthquake catalog and gives the alarming area in circular region for next five years [17, 18]. There are several studies conducted based on this algorithm and its advanced stabilized version i.e. M8S algorithm to forecast the seismic events of magnitude 5.5 and above [19].

The Three-Dimensional Pattern Informatics (PI) approach is also applied which aims at the forecast of earthquakes with natural and synthetic data sets [20]. Considering the regional seismic environment, the method efficiently characterizes the spatial and temporal seismic activity patterns with angular allusion occurred in the extent of associated space. This technique is the improved version of two-dimensional PI approach [21], in the sense that it resolves the vertically distributed seismic anomalies in the presence of complex tectonic structures. Moreover, it gives forecast by systematically analyzing the anomalous behaviors in the seismicity at regional level.

- Proposed System

This research work employs the known mathematical methods and computes all the seismic features in a bid to retain maximum information, which leads to sixty seismic features corresponding to every earthquake occurrence (Et). After acquiring maximum available seismic features, Maximum Relevancy and Minimum Redundancy (mRMR) based feature selection is applied to select most relevant feature having maximum information. Support Vector Regression (SVR) followed by Hybrid Neural Network (HNN) model and Enhanced Particle Swarm Optimization (EPSO) is applied to model relationship between feature vectors and their corresponding Et, thereby generating robust earthquake prediction model (SVR-HNN). The distinctive contributions of this research methodology is summarized as:

- i. Sixty seismic features, calculated on the principle of retaining maximum information.
- ii. A unique application of SVR in combination with HNN, on mRMR selected seismic features.

Hindukush, Chile and Southern California are three of the most active seismic regions in the world, thus considered in this study for studying problem of earthquake prediction. Earthquake prediction studies based on Artificial Neural Networks (ANN) have been performed on the considered seismic regions [9, 11, 15].

- Optional Functional Units

- Safety Confirmation

- You can post "Safety" or "Damaged" with name and phone number.
 - You can search Safety Confirmation by name or phone number.

- Prevention

- You can share the knowledge about disaster prevention.

- Voice navigation

- When "Voice" of the notification sound is selected, I inform you by voice when receiving an Earthquake Early Warning.

- Social Media

- User will have the share to social media option to share the notification to let people know about the disaster.

Chapter 4

Design

Systems design is the process of defining the architecture, components, modules, interfaces, and data for a system to satisfy specified requirements.

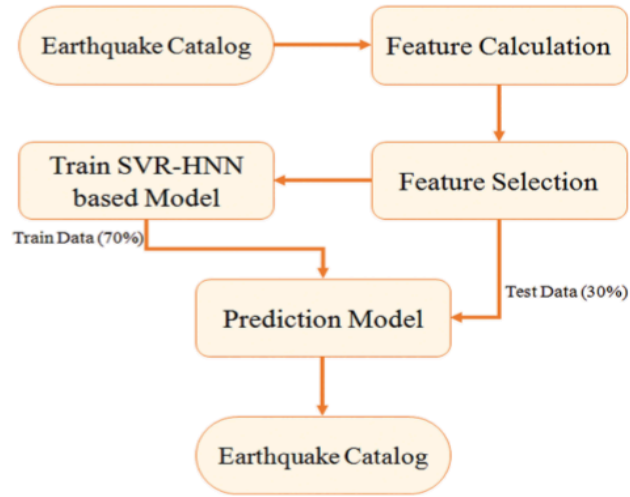
4.1 System Architecture

This application will be based on a three-tier architecture which includes database layer, application layer and the presentation layer. Database layer contains all the information about the type of data I will use in my model, application layer will give me a note on the processing of that data such as adding most relevant data and discarding the one which is not useful, data transformation, parameter tuning and next training a model on that given data will come in my application layer. Finally, I have my third layer which will deliver the results in the form of excel sheets, graphs and mobile application used to show output will be included in my presentation layer.

Database Layer:

This application will be a multi-step prediction model termed as (SVR-HNN). It is a combination of various machine learning techniques. Data set will be obtained for all the three regions and then divided into training and testing sets. For training and validation purposes, 70% of the data set will be selected, while testing will be performed on rest of 30% hold out data set. The final prediction results will be calculated separately for all the three regions that I will get on test data-set. The reason for performing separate training is that every region might have different properties and can be classified tectonically into different categories, such as thrusting tectonics, strike-slip tectonics and so forth. Therefore, every type of region possesses different behaviors and relations to the earthquakes. The data-set that I will use will be in the form of flat files such as excel sheets on every different region. The following fig.1 shows how to train and test my data.

Fig.1



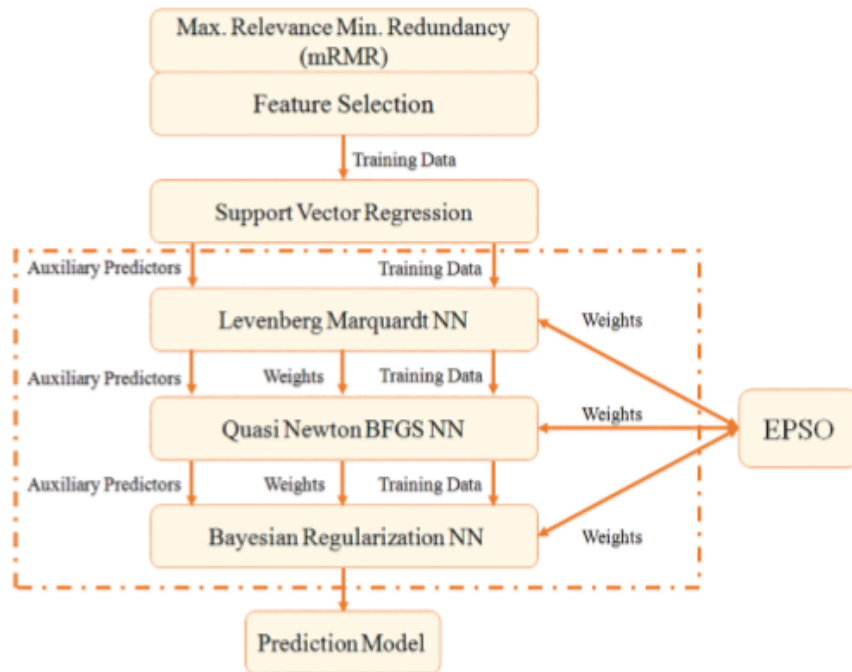
Application Layer:

The methodology I will use for calculating data set is a two-step model based on mRMR which mean maximum relevancy and minimum redundancy so after gathering data, the most relevant data is chosen for the prediction model to give results and the ones which will be of no more use for the model are also discarded. These checks are performed so only useful features are employed for earthquake prediction. The selected set of features are then passed to Support Vector Regression (SVR).

SVR is a machine learning technique that learns in a supervised manner. The model then generated through training of SVR, will give predictions about an estimated earthquake magnitude corresponding to the feature vectors. SVR will now transfer its knowledge to ANN which is the next level of prediction.

Next there will be three different layers of ANN in combination with EPSO which suggests input to one layer of ANN and its output used as an input to the next layer of ANN and so on. The reason for using output of one layer as an input to the other is so that the ANN does not start learning from scratch. EPSO here will be used for optimization techniques.

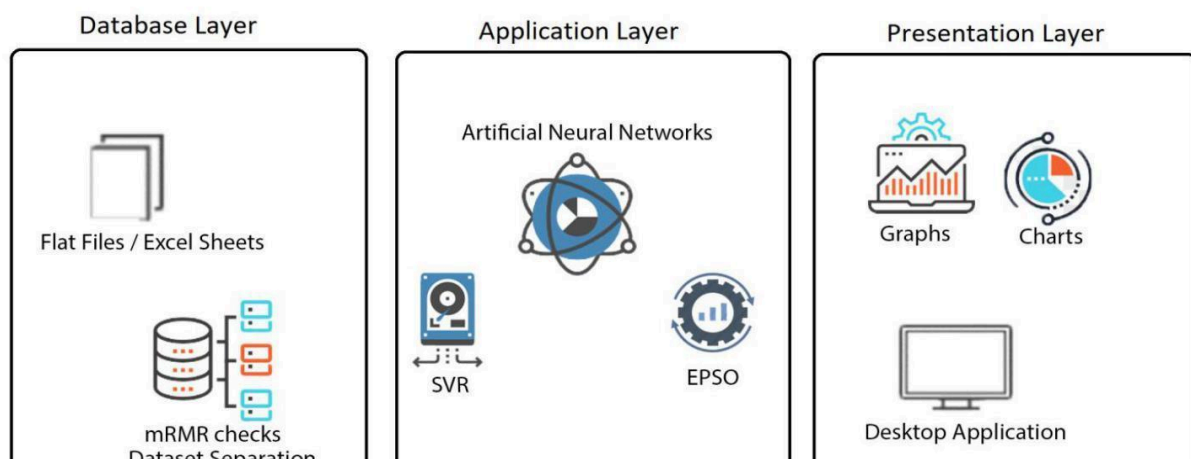
Fig.2



Presentation Layer:

After processing and training my model on the data-set I will represent the results in the form of graphs and excel sheets. So, this means the results will be in the form of flat files and graphs used to represent the percentages of the predicted results. Moreover, I will include my windows application which will be able to show these results graphically which will also provide a GUI interface. The following figure illustrates about my 3-tier application.

Fig.3

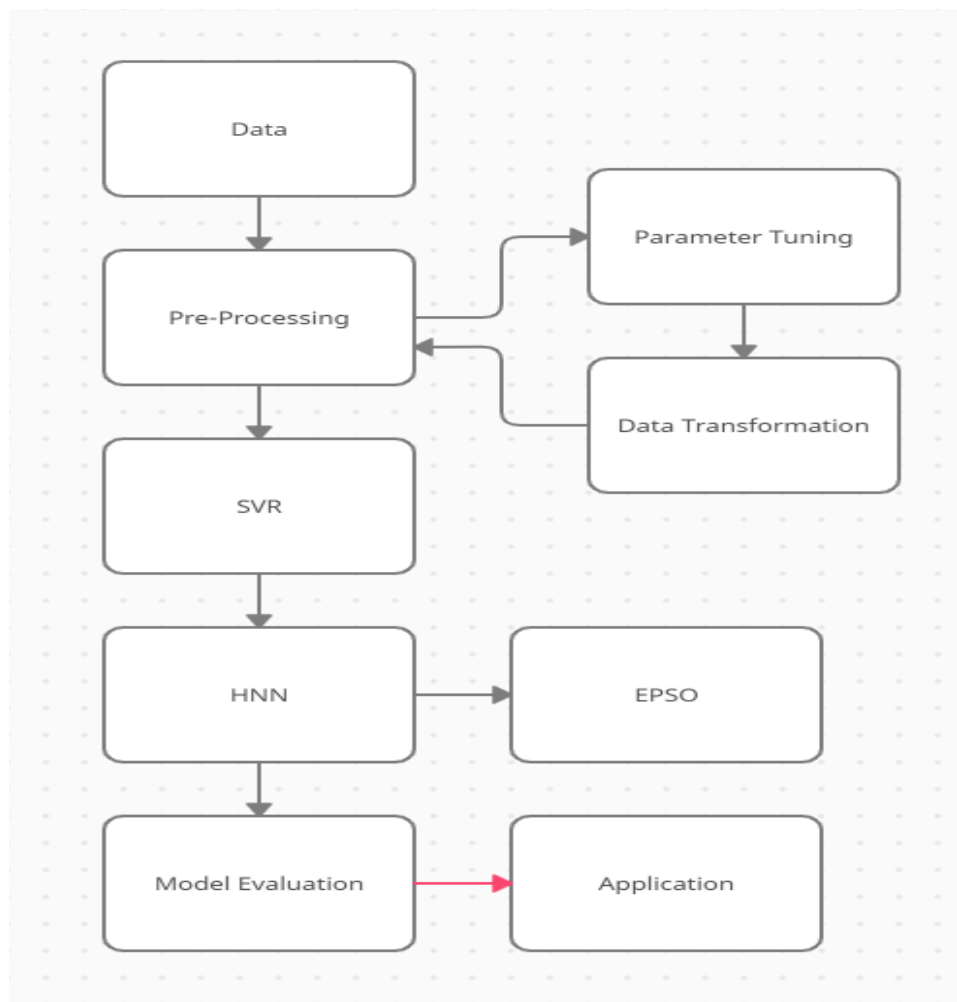


4.2 Design Constraints

The execution Time of the application is large and can take time to train execute and optimize using EPSO.

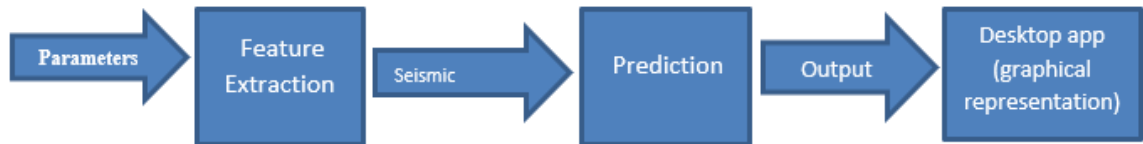
4.3 Design Methodology

Fig.4



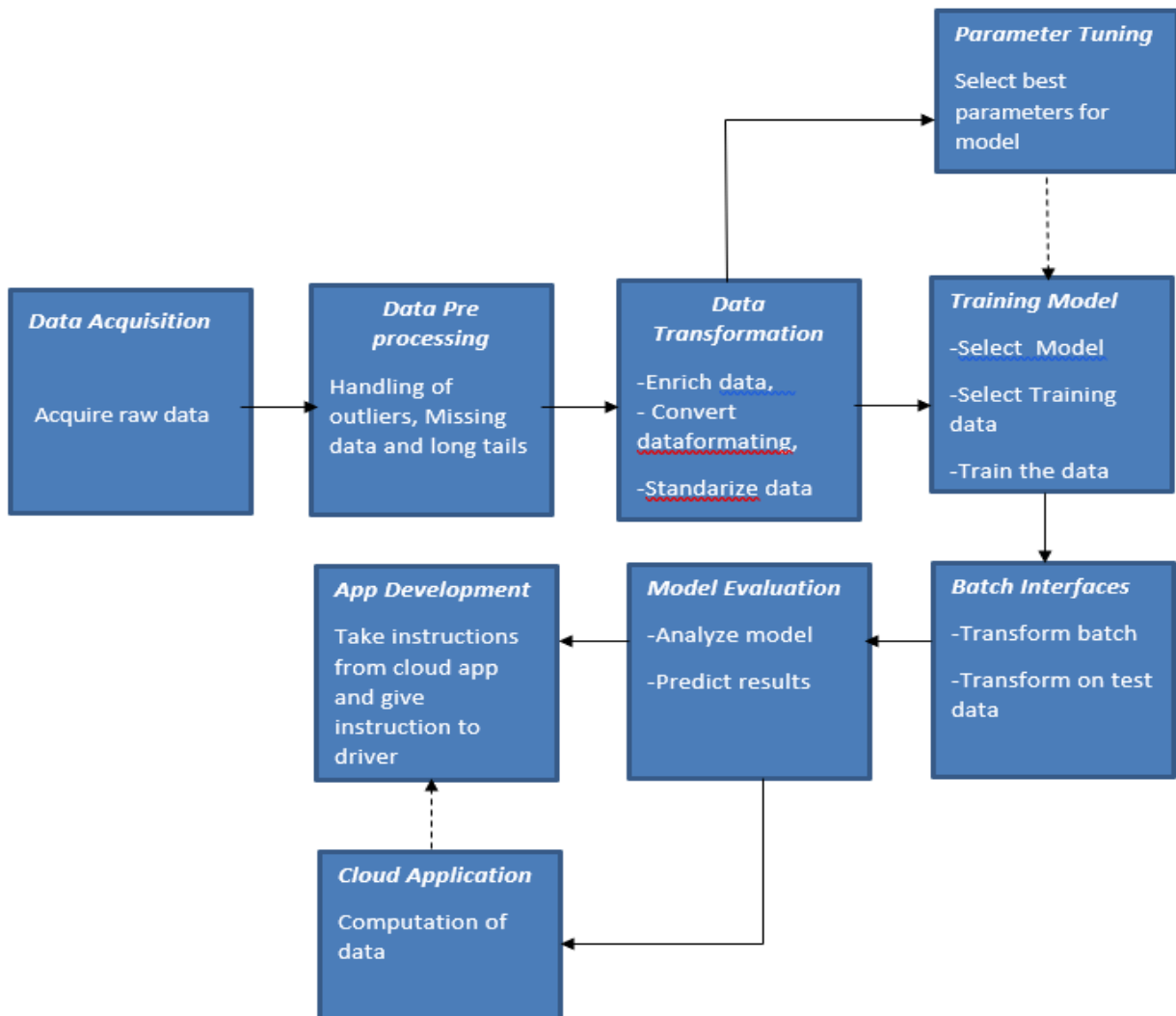
4.4 High Level Design

Fig.5



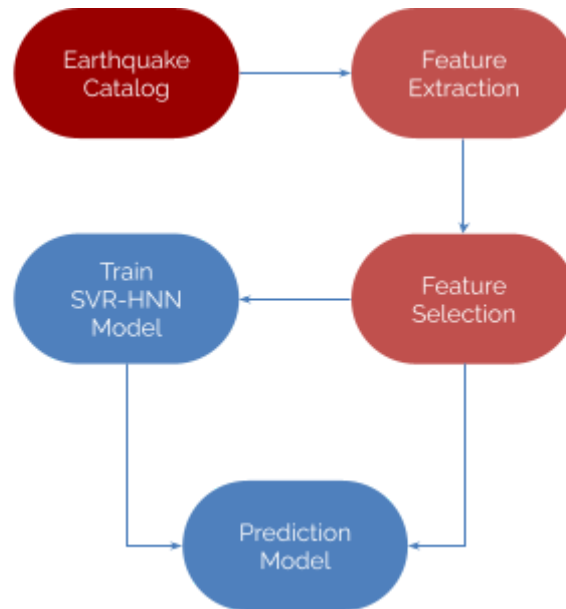
Step by step model implementation is following:

Fig.6



4.5 Low Level Design

Fig.7



4.6 Database Design

The database used here is firebase real-time database. Data is gathered from the official site <https://earthquake.usgs.gov/earthquakes/search/> for the 3 regions Chile, Hindukush and Southern California in which tectonic features are present and moreover the date, time place, latitude and longitude values of the earthquake and the magnitude of the earthquakes listed. Which is then accessed by the real-time database and from the database to the application to display the regions.

Firebase is a platform developed by Google for creating mobile and web applications. It was originally an independent company founded in 2011. In 2014, Google acquired the platform and it is now their flagship offering for app development.



Google Firebase is a Google-backed application development software that enables developers to develop iOS, Android and Web apps. Firebase provides tools for tracking analytics, reporting and fixing app crashes, creating marketing and product experiment.

Fig.8



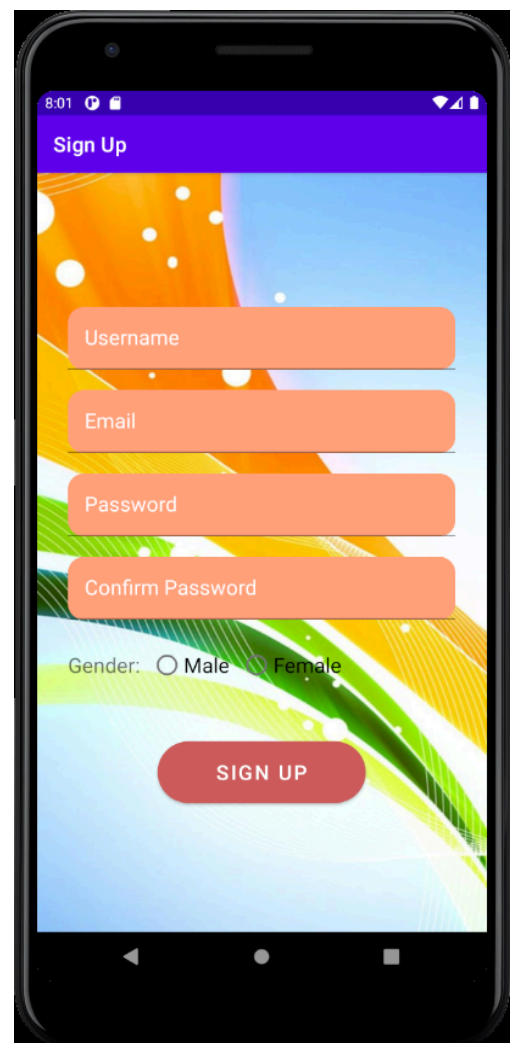
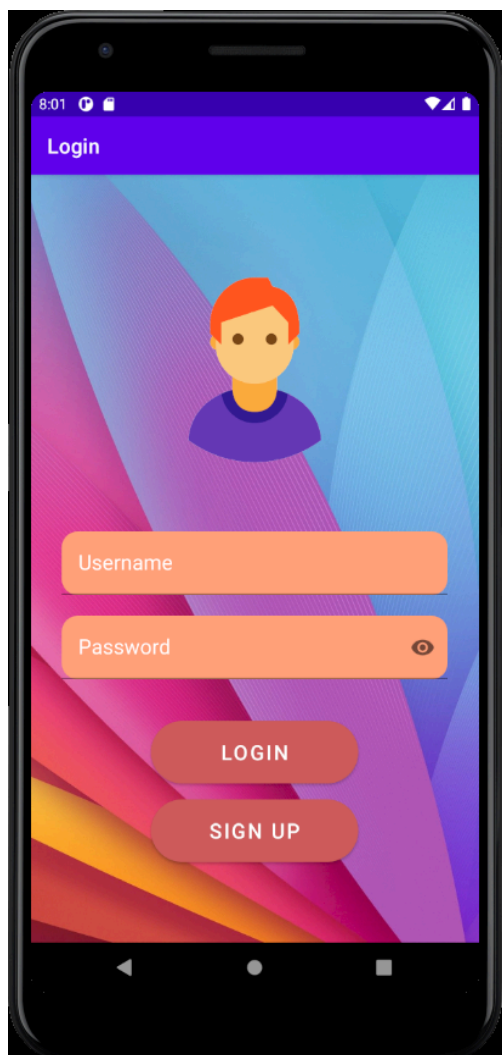
4.7 GUI Design

This project does not have a much-needed GUI as it does not require any fancy designed application. Requires only the GUI to display the map and mark the regions related to earthquakes and display information of earthquakes on the application.

4.8 External Interfaces

This Project can contain an android application which is connected to the project via firebase database and can be used to display the results graphically and the results can then be evaluated or discussed using the application but this application is optional and is not included within the scope of the system.

Android software development is the process by which applications are created for devices running the Android operating system. Google states that "Android apps can be written using Kotlin, Java, and C++ languages" using the Android software development kit.



Chapter 5

System Implementation

This is a multistep prediction model (SVR-HNN). It is a combination of various machine learning techniques with every technique complementing the other through knowledge acquired during learning. Thus, every step in this model is adding further improvements to the robustness therefore, resulting in a final improved version of prediction model. Dataset obtained for all the three regions, is divided into training and testing sets. For training and validation purposes, 70% of the dataset is selected, while testing is performed on rest of 30% hold out dataset. The prediction results obtained on test dataset for every region, separately. The configurations and setups arranged in order to train a model are kept same for all three regions' datasets. However, separate training has been performed for each region. The reason for separate training is that every region has different properties and can be classified tectonically into different categories, such as thrusting tectonics, strike-slip tectonics and so forth. Therefore, every type of region possesses different behaviors and relations to the earthquakes. Thus, separate training for every region is meant to learn and model the relationship between seismic features and earthquakes for that particular region. The proposed methodology includes the use of two step feature selection process. The features are selected after performing relevancy and redundancy checks, to make sure that only useful features are employed for earthquake prediction. The selected set of features are then passed to Support Vector Regression (SVR). The trend predicted by SVR is further used in combination with seismic features as input to the next stage of prediction model, i.e. ANN. The inclusion of SVR-output as a feature to ANN is to pass on the information learnt through SVR. After SVR, three different layers of neural networks are applied to the dataset in combination with Enhanced Particle Swarm Optimization (EPSO). The output of every ANN is used as an input to the next ANN in place of SVR-output along with feature set. The weight adjustments of each ANN layer is also passed to the next ANN, so that next ANN does not start learning from scratch. The purpose of including EPSO is to optimize the weights of ANN, which have tendency to get trapped in local minima. If during training ANN is stuck in local minima, EPSO plays a vital role in that scenario. The similar type of approach has also been used in other fields, such as wind power prediction with successful outcomes.

Chapter 6

System Testing and Evaluation

There are well-known performance metrics available for evaluation of binary classification results, such as sensitivity, specificity, positive predictive value or precision (P1), negative predictive value (P2), Matthews correlation coefficient (MCC) and R score. Whenever, predictions are made through an algorithm, it yields four categories of outputs, namely true positive (TP), false positive (FP), true negative (TN) and false negative (FN). TP and TN are the correct predictions made by the prediction model while FP and FN are the wrong predictions of the algorithm. Sensitivity (Sn) is the rate of true positive predictions out of all positive instances while specificity (Sp) is the rate of true negative predictions out among all negative instances. P1 is the ratio of correct positive predictions out of all the positive predictions made by prediction model, whereas P0 refers to the correct negative predictions made by algorithm out of all negative predictions. P1 inversely relates to false alarms, which refers higher P1 means lesser false alarms and vice versa. Similarly, R score and MCC are also proposed as a balanced measure for binary classification evaluation. These are calculated using all four basic measures (TP, FP, TN, FN) and vary between -1 and +1. The values approaching +1 correspond to perfect classification, while 0 refers to total random behavior of prediction algorithm and -1 relates to opposite behavior of classification model. These two performance measures can be considered as a benchmark measure for drawing comparison, because the four types of basic measures are also incorporated in them. Accuracy is a very general criterion for evaluation, used in many perspectives. It states the overall accurate predictions made by the algorithm.

The performance of SVR, HNN and EPSO are dependent on respective parameter selection. Thus, the collective performance obtained in SVR-HNN model is governed through selection of appropriate parameter values. The extensive experimentation is performed to empirically select the values for parameters, which obtain good results for each of the used algorithms (SVR, HNN, and EPSO). The proposed model has shown consistent performance for the three regions, which approves the appropriate selection of values for parameters. The ten simulation runs of SVR-HNN model show little variation, which strengthens the claim of appropriate selection of parametric values, leading to a stable earthquake prediction performance in three regions.

5.1. Results Achieved So Far

Accuracy achieved.

Fig.9

SVR

SVR_Contengency_test											
1x11 double											
1	2	3	4	5	6	7	8	9	10	11	
1	182	36	917	779	0.1894	0.9622	0.1894	0.5407	0.5742	0.2591	0.1326
2											
3											
4											

1st Hnn

Contengency_test1											
1x11 double											
1	2	3	4	5	6	7	8	9	10	11	
1	437	173	780	524	0.4547	0.8185	0.4547	0.5982	0.6358	0.3442	0.2157
2											
3											
4											

Last EPSO

Contengency_test1											
1x11 double											
1	2	3	4	5	6	7	8	9	10	11	
1	598	204	749	363	0.6223	0.7859	0.6223	0.6736	0.7038	0.5385	0.2912
2											
3											
4											

Results and states for training the Neural Network:

Fig.10

Performance

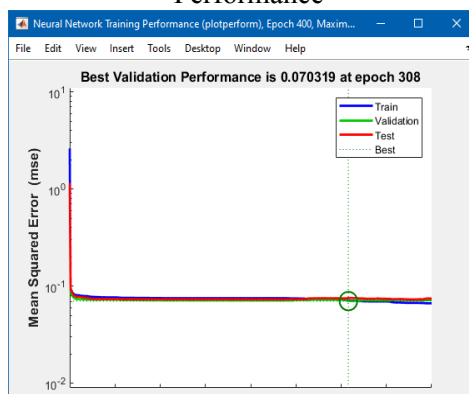
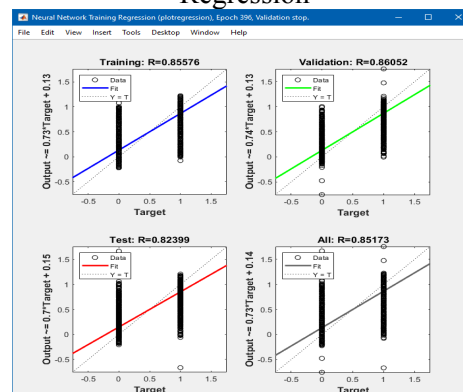


Fig.12

Fig.11

Regression



Training State

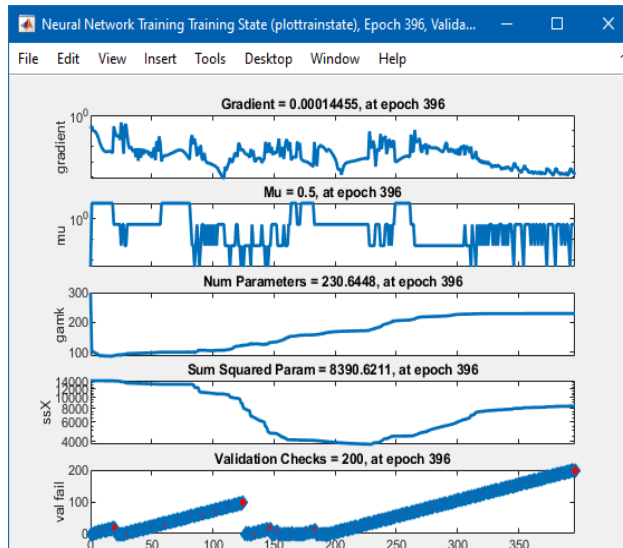


Fig.13

Bar Chart

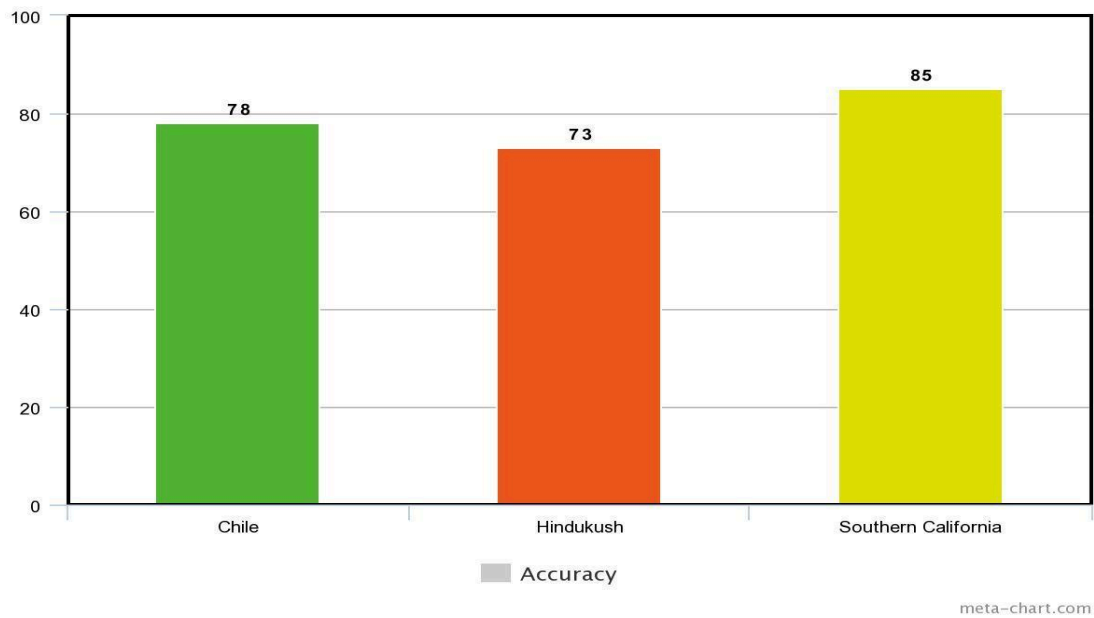


Fig.14

Application Displaying the regions where red zone and safe zone occurs.



Chapter 7

Conclusions

In this study, interdisciplinary research has been carried out for earthquake prediction through interaction of seismology-based earthquake precursors and computer science based computational intelligent techniques. A robust multilayer prediction model is generated in combination with the computation of maximum obtainable seismic features. Sixty seismic features are computed for Hindukush, Chile and Southern California regions. Separate feature selection is performed for every region through maximum relevance and minimum redundancy (mRMR) approach. The selected features are employed for training of an earthquake prediction model. The prediction model consists of Support Vector Regressor (SVR) followed Hybrid Neural Networks (HNN) and Enhanced Particle Swarm Optimization (EPSO). SVR provides an initial estimation for earthquake prediction which is passed to HNN as an auxiliary predictor in combination with features. Three different neural networks are further employed along with EPSO weight optimization. Thus SVR-HNN based prediction model is trained and tested successfully with encouraging and improved results for all three regions.

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